

Technology Stack

Component	Technology / Library
Language	Python 3.x
UI Framework	Streamlit
Data Handling	Pandas, NumPy
Data Cleaning	Pandas regex, str.extract, pd.to_numeric
Preprocessing	Scikit-Learn ColumnTransformer, MinMaxScaler, SimpleImputer
Feature Reduction	Scikit-Learn VarianceThreshold, SelectFromModel
Classification	RandomForestClassifier, DecisionTreeClassifier
Evaluation	train_test_split, StratifiedKFold, confusion matrix, classification
Ranking	Scikit-Learn cosine_similarity / weighted similarity score
Model Persistence	joblib for saved Random Forest and CART pipelines

Key Dependencies from requirements.txt: joblib==1.5.3 scikit-learn==1.8.0
pandas==3.0.0 numpy==2.4.2 scipy==1.17.0